This assignment will not be graded and is only for practice.

Multiple Select Question (MSQ)

1) Consider a function $f : \mathbb{R}^2 \to \mathbb{R}$ defined as: ***1 point***

$$f(x,y) = \begin{cases} \frac{2xy}{\sqrt{2(x^2+y^2)}} & \text{if } (x,y) = (0,0) \\ 1 & \text{if } (x,y) = (0,0). \end{cases}$$

Choose the set of correct options.

- [ ] $|x| \leq \sqrt{x^2+y^2}$ and $|y| \leq \sqrt{x^2+y^2}$.
- [ ] If $(x,y) \to (0,0)$, then $f(x,y) \to 0$.
- [ ] $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ exists.
- [ ] $\lim\limits_{(x,y)\to(0,0)} f(x,y) = 0$.
- [ ] $f(x,y)$ is continuous at the origin (0,0).

No, the answer is incorrect.

Score: 0

Accepted Answers:

$|x| \leq \sqrt{x^2+y^2}$ and $|y| \leq \sqrt{x^2+y^2}$.
If $(x,y) \to (0,0)$, then $f(x,y) \to 0$.
$\lim\limits_{(x,y)\to(0,0)} f(x,y)$ exists.
$\lim\limits_{(x,y)\to(0,0)} f(x,y) = 0$.

2) Consider the following functions ***1 point***

$$f(x,y) = (\sqrt{1-x}, \sqrt{y})$$

$$g(x,y) = \sin((x^2-1)^2 + y^4)$$

$$h(x,y) = \frac{x}{x^2+y^2}$$

$$s(x,y) = x^2 e^{y^2}$$

Which of the following is true?

- [ ] Domain of $f = \{(x, y) \mid x \leq 1, y \geq 0\}$
- [ ] Domain of $h = \mathrm{R}^2$
- [ ] $g \circ f(x, y) = \sin(x^2 - 1 + y^2)$
- [ ] $((g \circ f) \times h)(x, y) = \frac{\sin(x^2+y^2)}{x^2+y^2}$
- [ ] $\lim_{(x,y) \to (0,0)} ((g \circ f) \times h + s)(x, y) = 0$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Domain of $f = \{(x, y) \mid x \leq 1, y \geq 0\}$
$\lim_{(x,y) \to (0,0)} ((g \circ f) \times h + s)(x, y) = 0$

3) Match the functions of two variables in Column A with their graphs given in Column B. ***1 point***

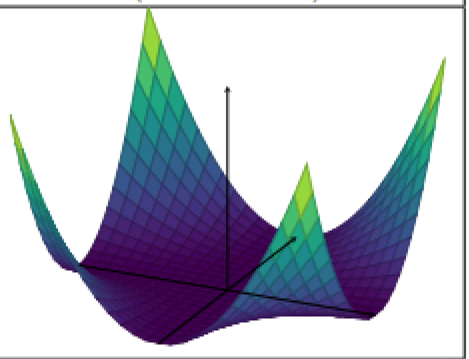
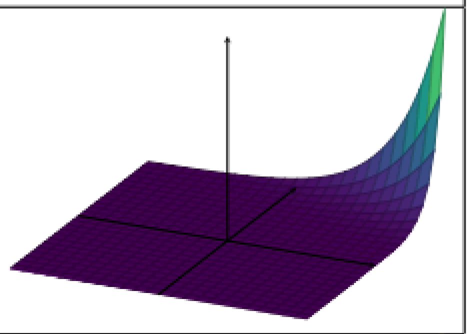
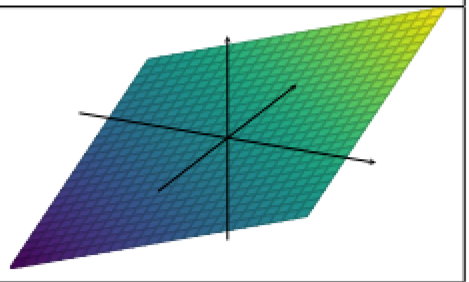
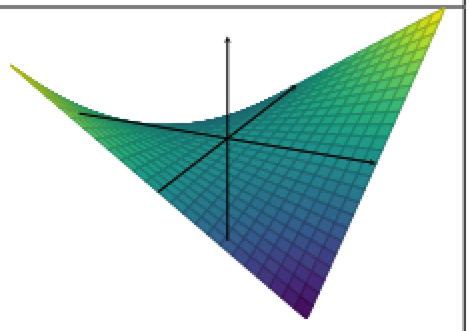
	Function of two variables (Column A)		Graph of the function (Column B)
i)	$f(x, y) = x + y$	1)	
ii)	$f(x, y) = xy$	2)	
iii)	$f(x, y) = x^2y^2$	3)	
iv)	$f(x, y) = 5^{2x+4y}$	4)	

Table: M2W9P1

- i → 4, ii → 3
- i → 3, ii → 4
- iii → 1, iv → 2
- iii → 2, iv → 1

No, the answer is incorrect.
Score: 0

Accepted Answers:

i → 3, ii → 4
iii → 1, iv → 2

4) Consider two functions $f : \mathbb{R}^2 \to \mathbb{R}^2$ and $g : \mathbb{R}^2 \to \mathbb{R}$ defined as: ***1 point***

$$f(x,y) = (x^2, y^2)$$

and

$$g(x,y) = \begin{cases} \frac{x^2-y^2}{x^2+y^2} & \text{if } (x,y) = (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Which of the following option(s) is (are) true?

- [ ] $g \circ f(x,y) = \begin{cases} \frac{x^2-y^2}{x^2+y^2} & \text{if } (x,y) = (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$
- [ ] $g \circ f(x,y) = \begin{cases} \frac{x^4-y^4}{x^4+y^4} & \text{if } (x,y) = (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$
- [ ] $\lim\limits_{(x,y)\to(0,0)} g(x,y)$ exists.
- [ ] $\lim\limits_{(x,y)\to(0,0)} g \circ f(x,y)$ exists.
- [ ] $\lim\limits_{x\to 0}\lim\limits_{y\to 0} g \circ f(x,y) = \lim\limits_{y\to 0}\lim\limits_{x\to 0} g \circ f(x,y)$
- [ ] $\lim\limits_{x\to 0}\lim\limits_{y\to 0} g(x,y) = 1$
- [ ] $\lim\limits_{y\to 0}\lim\limits_{x\to 0} g \circ f(x,y) = -1$
- [ ] $\lim\limits_{(x,y)\to(1,1)} g(x,y)$ does not exist.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$$g \circ f(x,y) = \begin{cases} \frac{x^4-y^4}{x^4+y^4} & \text{if } (x,y) = (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

$$\lim_{x\to 0}\lim_{y\to 0} g(x,y) = 1$$

$$\lim_{y\to 0}\lim_{x\to 0} g \circ f(x,y) = -1$$

Numerical Answer Type (NAT)

5) Consider a function $f(x,y) = xy$. If the directional derivative of f at point (a,b), in direction of (1,-1) and (1,1) are 1 and 5 respectively. Then find the value $\frac{2a+b}{\sqrt{2}}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 7

1 point

6) Consider the following statements:

Statement 1: Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function such that $\lim\limits_{x\to a}\lim\limits_{y\to b} f$ exists, where $a, b \in \mathbb{R}$. Then $\lim\limits_{(x,y)\to(a,b)} f$ exists.

Statement 2: Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function such that $\lim\limits_{y\to b}\lim\limits_{x\to a} f = f(a,b)$, where $a, b \in \mathbb{R}$. Then f is continuous.

Statement 3: Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function such that $\lim\limits_{x\to a}\lim\limits_{y\to b} f = \lim\limits_{y\to b}\lim\limits_{x\to a} f = f(a,b)$, where $a, b \in \mathbb{R}$. Then f is continuous.

Statement 4: Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function such that $f_x(a,b)$ exists. Then f is continuous.

Statement 5: Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function such that $f_y(a,b)$ exists. Then f is continuous.

How many statements are correct?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: String) 0

1 point

7) Consider the following two functions $f_1, f_2 : \mathbb{R}^2 \to \mathbb{R}$ defined as:

$$f_1(x, y) = xy + 2x$$
$$f_2(x, y) = x^2 + 2y + 3x^2y.$$

Define a matrix A as follows:

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x}(3,0) & \frac{\partial f_2}{\partial x}(1,1) \\ \frac{\partial f_1}{\partial y}(3,2) & \frac{\partial f_2}{\partial y}\left(\sqrt{\frac{7}{3}}, 2\right) \end{bmatrix}.$$

What will be the value of $det(A)$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -6

1 point

Comprehension Type Question

An earthworm is crawling about on a sunny day. We draw coordinates axes on the ground where it is crawling and find that the body temperature of the earthworm at a point (x, y) is given by the function:

$$T(x, y) = 2x^2 + 3xy + y^2$$

Use this information to answer questions 8,9, and 10.

8) Which of the following option(s) is (are) true? ***1 point***

- [ ] $\lim\limits_{(x,y)\to(1,1)} T(x, y) = \lim\limits_{x\to 1}\lim\limits_{y\to 1} T(x, y)$
- [ ] $T(x, y)$ is continuous at (1,1)
- [ ] $T(x, y)$ is a linear function.

- [ ] $T(cx, cy) = c\,T(x, y)$.
- [ ] If the earthworm approaches the point $(1, 2)$ from any direction, then the body temperature of the earthworm approaches 12 units.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lim_{(x,y)\to(1,1)} T(x, y) = \lim_{x\to 1}\lim_{y\to 1} T(x, y)$

$T(x, y)$ is continuous at (1,1)

If the earthworm approaches the point $(1, 2)$ from any direction, then the body temperature of the earthworm approaches 12 units.

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9) If the gradient of $T(x, y)$ is (25, 18) at a point, then find the body temperature of the earthworm at that point.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 77

1 point

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10) Which of the following option(s) is (are) true? ***1 point***

- [ ] If the earthworm moves parallel to the $X-$axis, then rate of change of its body temperature at the point (a, b) is given by $4a + 3b$.
- [ ] If the earthworm moves parallel to the $Y-$axis, then rate of change of its body temperature at the point (a, b) is given by $2a + 3b$.
- [ ] If the earthworm moves parallel to the straight line joining the origin and the point $(2, 3)$, then rate of change of its body temperature at the point $(1, 1)$ is given by $\frac{29}{\sqrt{13}}$.
- [ ] If the earthworm moves parallel to the straight line joining the origin and the point $(0, 1)$, then rate of change of its body temperature at the point $(1, 2)$ is given by 10.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If the earthworm moves parallel to the $X-$axis, then rate of change of its body temperature at the point (a, b) is given by $4a + 3b$.
If the earthworm moves parallel to the straight line joining the origin and the point $(2, 3)$, then rate of change of its body temperature at the point $(1, 1)$ is given by $\frac{29}{\sqrt{13}}$.

Your score is: 0/10